

Design and Simulation of Silicon Photonic Interferometers

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Six TE interferometers are simulated to determine free spectral range and waveguide group index.

I Theory

$$n_g = n_p - \lambda \frac{dn_p}{d\lambda}, \quad f_i = \frac{c}{\lambda_i} \quad (1)$$

$$\Delta f = \frac{1}{N-1} \sum |f_{i+1} - f_i|, \quad \Delta\lambda \approx \frac{\lambda^2 \Delta f}{c} \quad (2)$$

$$n_{g,MZI} = \frac{c}{\Delta L \Delta f}, \quad n_{g,MM} = \frac{c}{2\Delta L \Delta f} \quad (3)$$

ΔL is the arm-length difference; L is the straight FP cavity length. $c = 299,792,458$ m/s. N is the number of selected resonances. FP and MFP use the complex grating response in Section III.

II Design and simulation

Device	Length	Bragg mirrors
MZI 30 μm	$\Delta L = 29.998 \mu\text{m}$	—
MZI 150 μm	$\Delta L = 150.002 \mu\text{m}$	—
MM 100 μm	$\Delta L = 100 \mu\text{m}$	300 / 300 periods, 200 nm corrugation
FP 150 μm	$L = 149.999 \mu\text{m}$	100 / 100 periods, 50 nm corrugation
FP 250 μm	$L = 249.999 \mu\text{m}$	100 / 100 periods, 50 nm corrugation
MFP 100 μm	$L = 100 \mu\text{m}$	10 / 300 / 300 periods, 200 nm corrugation

Silicon strip: 500 nm width, 220 nm thickness; TE polarization; 5 μm bend radius. Bragg period: 317 nm. Directional coupler: 200 nm gap, 10 μm coupling length.

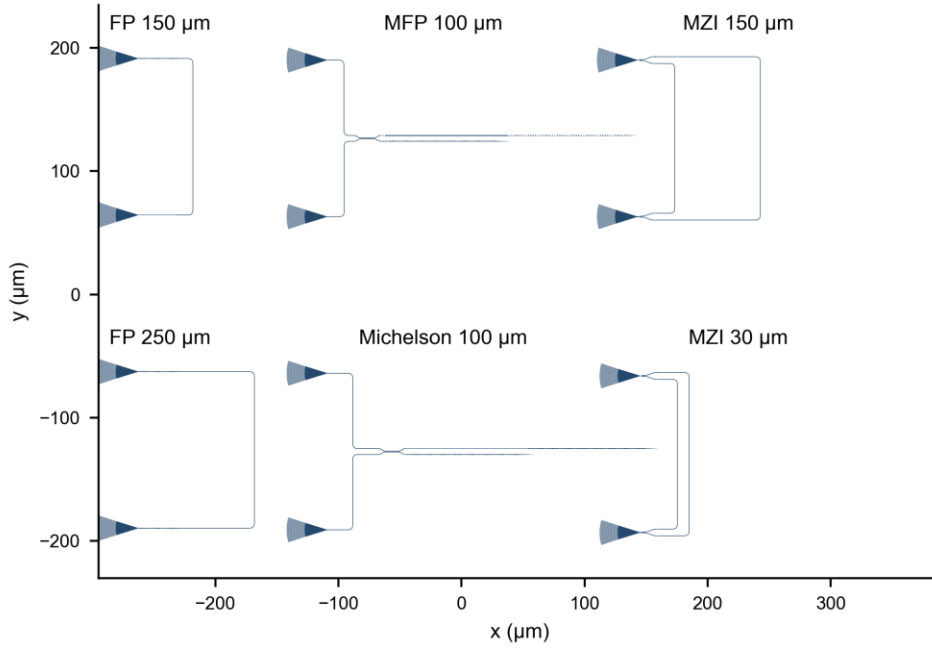


Figure 1 EBeam_weizhu.gds layout. Six input–output pairs with 127 μm grating-coupler spacing. Optical connectivity: 39 matched pin pairs.

PDK strip model at 1550 nm: $n_p = 2.44553$, $n_g = 4.19088$. The separate MODE export wg.mat gives $n_p = 2.4462$ and $n_g = 4.2036$ at 1550 nm.

Simulation results

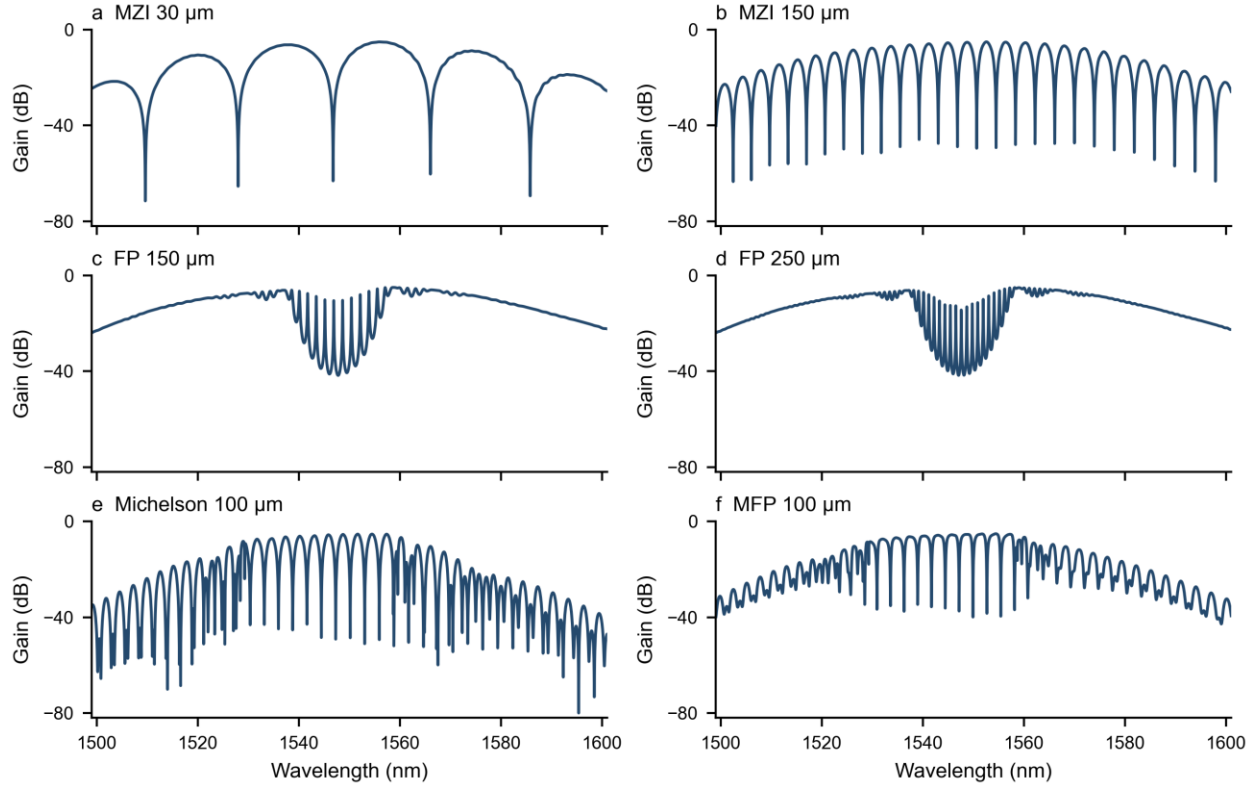


Figure 2 Saved INTERCONNECT gain spectra. 5,000 samples per device over 1499.02–1601.00 nm.

Device	FSR nm	FSR GHz	Group index
MZI 30 μm	19.03	2384.3	4.192
MZI 150 μm	3.81	476.6	4.193
MM 100 μm	2.86	357.5	4.193
FP 150 μm	1.74	217.3	4.196
FP 250 μm	1.08	135.1	4.192
MFP 100 μm	2.75	343.8	4.188

FSR uses consecutive FP transmission maxima and MZI, MM and MFP minima. Selected features: 1542–1555 nm; MZI 30: 1520–1575 nm. FP and MFP indices are fitted with the nominal PDK grating model.

Group index calculation

Device	Calculation	n_g
MZI 30	$299792458 / (29.998 \times 10^{-6} \times 2384.2721 \times 10^9)$	4.192
MZI 150	$299792458 / (150.002 \times 10^{-6} \times 476.6223 \times 10^9)$	4.193
MM 100	$299792458 / (2 \times 100 \times 10^{-6} \times 357.4784 \times 10^9)$	4.193

$$n_{g,FP\ pair} \approx \frac{c}{2 \times 100 \times 10^{-6}} \left[\frac{1}{135.0940 \times 10^9} - \frac{1}{217.3032 \times 10^9} \right] = 4.198 \quad (4)$$

FP pair estimate assumes equal mirror delay over the selected frequency windows. FSR measures group delay; the absolute phase index n_p is supplied by the waveguide model.

Data files: TE_MZI_30um.txt, TE_MZI_150um.txt, TE_MM_100um.txt, TE_FP_150.txt, TE_FP_250.txt and TE_MFP_100um.txt.

III FP and MFP calculation

$u = \exp(i\beta L)$. $\beta(f) = \beta_0 + 2\pi n_g(f - f_0)/c$. r and t are complex grating reflection and transmission coefficients.

$$H_{FP} = \frac{t^2 u}{1 - r^2 u^2} \quad (5)$$

$$r_{cav} = r_f + \frac{t_f^2 r_b u^2}{1 - r_f' r_b u^2} \quad (6)$$

MFP: f = front grating; b = rear grating; r_f' = reflection from the cavity side. The front grating is $3.17 \mu\text{m}$ long; the straight cavity is $100 \mu\text{m}$.

$$P_{FP} \propto |H_{FP}|^2, \quad P_{MFP} \propto |r_{cav} + r_{ref}|^2 \quad (7)$$

Grating model	FP mirrors	MFP front rear reference
Period	317 nm	317 nm
Corrugation	50 nm	200 nm
Periods	100 / 100	10 / 300 / 300
Bragg wavelength	1547.7045 nm	1543.8124 nm
Coupling κ	$+75375.25 \text{ m}^{-1}$	-159056 m^{-1}
Loss	590 dB/m	590 dB/m

Complex r and t use the uniform coupled-mode grating response with these nominal EBeam model parameters. n_g is fitted freely to all 638 samples at 1542–1555 nm, with a constant phase offset, smooth gain envelope and small power floor.

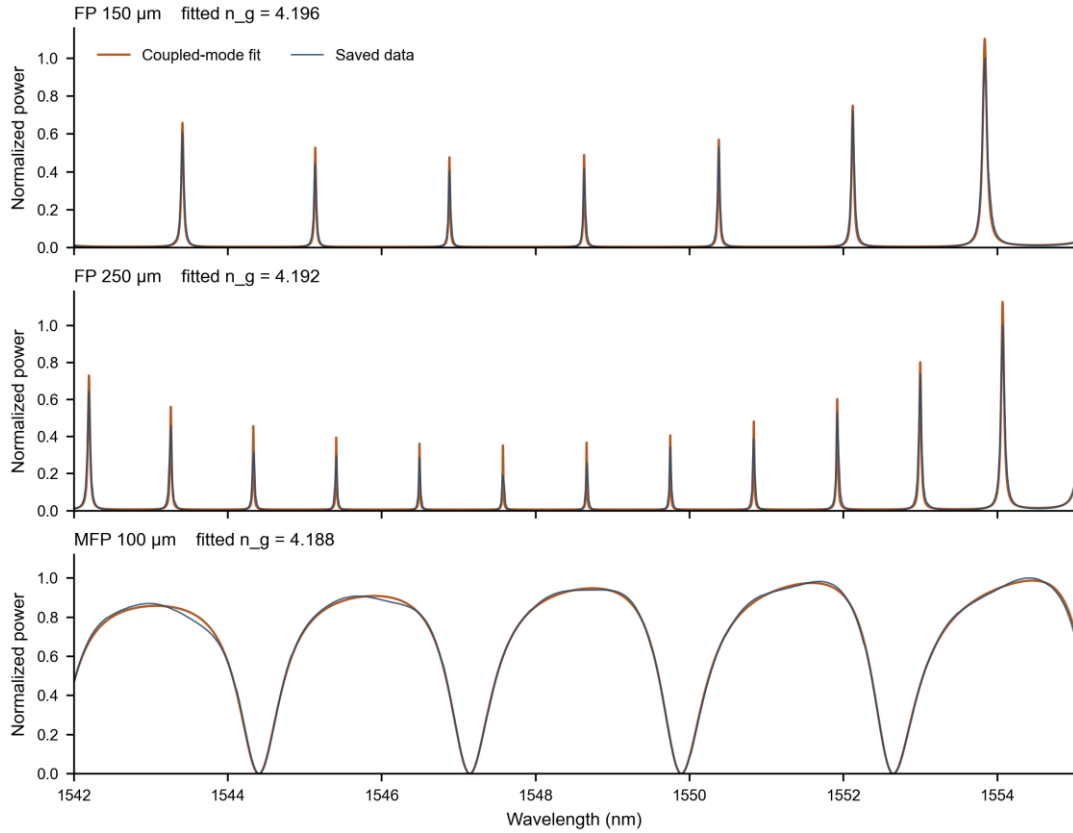


Figure 3 Saved power and fitted response. RMS residual relative to the maximum saved power in this window: FP 150, 1.29%; FP 250, 1.93%; MFP 100, 1.05%.

Model sources: installed EBeam ebeam_bragg_te1550.ice and WaveGuideTETMStrip,w=500,h=220.txt; Ansys INTERCONNECT Bragg Grating WBG element documentation.